

KENYA AI CHALLENGE 2026

AgriFin × AI — Project Breakdown & Judge Simulation

Prepared for: Team of 5 Builders

Challenge Partner: Mercy Corps AgriFin

Final Sprint: 27–28 June 2026, Nairobi

Four Projects. Every Angle. You Decide.

How to Use This Document

This document gives you a complete breakdown of all four project ideas for the Kenya AI Challenge 2026. For each project you will find:

Section	What it covers
Overview	What the project is and what it does in plain language
The Problem It Solves	The specific AgriFin pain point it addresses
How It Works	Technical and product architecture, step by step
Tech Stack	Which sponsor and non-sponsor tools are used and why
Strengths	Why judges and partners would love it
Weaknesses	Honest risks and gaps — do not ignore these
Judge Q&A Simulation	The toughest questions a judge will ask, with model answers
Verdict	A direct assessment of win probability and team fit

Read all four sections before making your decision. The judge Q&A section is especially important — these are the questions that have eliminated otherwise strong teams in past challenges.

Quick Comparison at a Glance

Project	Core Problem	Primary User	Technical Depth	Demo Power	Commercial Value	Win Probability
01 — Credit Scoring	Credit invisibility	Lenders	★★★★★	★★★★☆	★★★★★	★★★★☆
02 — Input Financing Assistant	Info asymmetry	Farmers	★★★★☆	★★★★★	★★★★☆	★★★★★
03 — Matchmaking Graph	Market fragmentation	Both	★★★★☆	★★★★☆	★★★★☆	★★★★☆
04 — Collection Agent	Loan defaults / NPLs	Lenders	★★★★☆	★★★★☆	★★★★★	★★★★☆

PROJECT 01 | FARM CREDIT SCORING AGENT

"Making invisible farmers visible to lenders"

Overview

Smallholder farmers in Kenya lack formal financial histories, making them "credit invisible" to traditional lenders. This project builds an AI-powered credit scoring engine that constructs alternative credit profiles using non-traditional data sources — satellite imagery, seasonal crop patterns, mobile money (M-Pesa) transaction signals, weather data, and agronomic inputs. The result is a score that lenders can trust and farmers can grow.

The Problem It Solves

Mercy Corps AgriFin has identified that the biggest blocker to agricultural finance is not the absence of loan products — it is the inability of lenders to assess risk for smallholder farmers. Most farmers have no payslip, no bank statement, no collateral. Banks and microfinance institutions pass on them entirely. This project directly solves that gap.

How It Works — Step by Step

1. Data ingestion layer: pulls in satellite NDVI (crop health) data, M-Pesa transaction patterns, weather station data, and farmer-reported input usage.
2. Neo4j graph layer: maps relationships between farmers, their crops, input suppliers, off-takers, and past loan records — revealing trust networks invisible in flat data.
3. AI scoring model: trained on proxy datasets (e.g., Mercy Corps AgriFin historical data) to produce a composite creditworthiness score.
4. Lender dashboard: a clean interface where financial institutions query a farmer ID and receive a structured credit report with score, explanation, and suggested loan ceiling.
5. Farmer-facing view: farmers can see their own score, understand what drives it, and get recommendations to improve it.

Tech Stack & Sponsor Tool Usage

- Neo4j — core graph database mapping farmer-lender-crop networks
- Featherless LLMs — natural language explanation of credit scores
- Lovable — rapid UI build for both lender dashboard and farmer portal
- Masumi — optional payment layer if loan disbursement is integrated
- Python/FastAPI — backend scoring engine

✓ Strengths	⚠ Weaknesses
Directly attacks the #1 AgriFin pain point identified by Mercy Corps	Requires real or realistic proxy data — hard to source in 6 weeks without partner access

Neo4j is a perfect fit — graph relationships are what make alternative credit data powerful	Risk of building a "demo with fake data" that judges see through
High technical depth impresses judges with engineering backgrounds	Complex to explain clearly to non-technical judges in a short pitch
Scalable: the model can be trained on more data post-challenge	Regulatory angle (credit scoring is regulated in Kenya) may invite tough questions
Addresses both sides of the market: lenders AND farmers	Needs an ML/data science person on the team
Strong follow-on commercial pathway — fintechs will pay for this	

Judge Q&A Simulation — Toughest Questions

These are the questions a judge from Mercy Corps AgriFin, a technical panel member, or a commercial investor would ask. Read every answer carefully — these are where challenges are won or lost.

Q1: "Where does your training data come from?"

A: We used publicly available NDVI satellite datasets, anonymised M-Pesa transaction patterns from published research, and synthetic farmer profiles modelled on Mercy Corps AgriFin published case studies. With partner access post-challenge, we would retrain on real data.

Q2: "How is this different from existing credit bureaus like Metropol or CRB Africa?"

A: Existing bureaus rely on formal credit history — which 80% of smallholder farmers do not have. Our model works with zero prior credit history, using agronomic and behavioural signals instead.

Q3: "Is this legal? Credit scoring is regulated."

A: We are building a decision-support tool, not a regulated credit bureau. Final lending decisions stay with the institution. We comply with Kenya's Data Protection Act 2019 by anonymising all personal data.

Q4: "Can farmers game the score?"

A: The model uses multiple independent signals. Gaming one (e.g., inflating input purchases) does not move the score significantly. The graph layer also detects anomalous patterns.

Q5: "What is your go-to-market after the challenge?"

A: Partner with one AgriFin-connected MFI or SACCO for a pilot. The graph and scoring engine are the moat — not easy to replicate quickly.

Verdict

HIGH AMBITION, HIGH REWARD — Best for a team with data science capability. If you can source credible data, this is a top-2 finalist project. Without good data, it risks feeling like a prototype demo.

PROJECT 02 | AGRI-INPUT FINANCING ASSISTANT

"A multilingual AI guide that walks farmers to their first loan"

Overview

Millions of Kenyan smallholder farmers know they need financing for seeds, fertiliser, and other inputs — but navigating the landscape of loan products, eligibility criteria, repayment schedules, and application forms is overwhelming, especially in English. This project is a conversational AI assistant available in Swahili and English that guides farmers through the entire input financing journey: understanding what is available, checking eligibility, simulating repayment, and initiating applications with AgriFin partner institutions.

The Problem It Solves

The gap is not just credit availability — it is information asymmetry. Farmers do not know what products exist, what they qualify for, or how repayment works in their context (post-harvest cash flows, drought risk, etc.). Financial institutions spend enormous resources on farmer education and last-mile outreach. This assistant automates that gap.

How It Works — Step by Step

6. Conversational interface: farmer types or speaks in Swahili/English. The assistant responds naturally, asking clarifying questions about farm size, crops, location, and financial history.
7. Product matching engine: against a database of AgriFin partner loan products, the assistant surfaces the 2-3 best-fit options for that farmer's profile.
8. Repayment simulator: farmer inputs expected yield and price — assistant shows exactly what repayment looks like across harvest cycles.
9. Application initiation: the assistant pre-fills an application form and either submits it to the partner institution or generates a ready-to-submit document.
10. Masumi integration: optional payment or deposit step to lock in the application or pay a processing fee via the assistant itself.

Tech Stack & Sponsor Tool Usage

- Featherless open-source LLMs — Swahili/multilingual conversational AI (models like Aya or mT5 fine-tuned for Swahili)
- Lovable — rapid frontend UI, mobile-first design
- Masumi — in-conversation payment/deposit trigger
- Node.js/Express or Python backend — product database and matching logic
- Neo4j — optional: graph of farmer profiles, loan products, and eligibility paths

✓ Strengths	⚠ Weaknesses
Most demo-able project — judges can literally talk to it in real time Swahili support is a major differentiator that signals real user empathy Solves a real, documented problem — information asymmetry in agricultural finance Works well even with synthetic product data during the challenge Lovable makes the UI fast to build — team can focus on the AI quality Easiest to explain to non-technical judges: "It is a WhatsApp-style bot that helps farmers apply for loans" Mercy Corps AgriFin can immediately envision deploying this with their network Strong on all 5 judging dimensions simultaneously	Swahili LLM quality may be inconsistent — open-source multilingual models are not perfect Without real loan product data from AgriFin partners, the matching engine uses dummy data Risk of being perceived as "just a chatbot" by technical judges if not built with depth Needs robust prompt engineering to avoid hallucinating loan terms or eligibility criteria WhatsApp integration (the real channel) is not buildable in the sprint — the demo is web-only

Judge Q&A Simulation — Toughest Questions

These are the questions a judge from Mercy Corps AgriFin, a technical panel member, or a commercial investor would ask. Read every answer carefully — these are where challenges are won or lost.

Q1: "Why not just use WhatsApp? That is where farmers actually are."

A: Our web demo proves the core AI works. WhatsApp Business API integration is our first post-challenge deployment step — we have already scoped the integration. The intelligence layer is what we are validating here.

Q2: "What happens if the LLM gives wrong loan terms?"

A: We use a retrieval-augmented approach — the LLM only answers from a verified product database, never from its own knowledge about loan terms. Hallucination is structurally prevented for financial facts.

Q3: "How is this different from a bank's existing USSD menu?"

A: USSD menus are rigid scripts. Our assistant understands natural language, handles follow-up questions, explains trade-offs between products, and simulates personalised repayment — none of which USSD can do.

Q4: "Can this work without internet? Most rural areas have poor connectivity."

A: The web interface requires data, yes. However, SMS/USSD fallback and lightweight offline caching are roadmap items. During the challenge we focus on proving the AI core works.

Q5: "What stops a big bank from building this in 3 months?"

A: Banks move slowly. The AgriFin-specific product knowledge, Swahili fine-tuning, and integration with Mercy Corps' partner network are moats. We are also open to white-labelling to any institution.

Verdict

HIGHEST WIN PROBABILITY — Best overall balance of impact, feasibility, demo quality, and judge appeal. Recommended for most team compositions. The risk is in LLM quality — test Swahili models early.

PROJECT 03 | FARMER-LENDER MATCHMAKING GRAPH

"A Neo4j-powered marketplace connecting the right farmer to the right loan"

Overview

Kenya has dozens of agricultural finance products across MFIs, SACCOs, commercial banks, and agri-input companies — but no centralised intelligence layer that matches a specific farmer to the product best suited to their situation. This project builds that layer using Neo4j as a graph database to model the relationships between farmers, crops, risk profiles, geographic zones, seasons, and loan products. An AI layer on top surfaces optimal matches, explains why, and guides both farmers and loan officers through next steps.

The Problem It Solves

Both sides of the market are inefficient. Farmers waste time applying for products they do not qualify for. Lenders waste resources processing ineligible applicants. The matching problem is inherently a graph problem — eligibility depends on interconnected attributes, not flat rules.

How It Works — Step by Step

11. Graph schema design: nodes for Farmer, Crop, Region, Season, LoanProduct, Lender, RiskCategory. Edges represent relationships like "grows", "qualifies_for", "historically_repaid", "operates_in".
12. Farmer onboarding: a simple form or conversational flow captures farmer attributes and populates the graph.
13. Match algorithm: Neo4j Cypher queries traverse the graph to find loan products where all eligibility conditions are met, ranked by fit score.
14. Explanation layer: Featherless LLM generates a plain-language explanation of why each match was returned and what the farmer needs to do to qualify for better products.
15. Loan officer portal: lenders can query the graph for farmers in their region who match their product criteria — reversing the direction of the match.

Tech Stack & Sponsor Tool Usage

- Neo4j — the entire core of the project; graph traversal is the intelligence
- Featherless LLMs — natural language explanations and conversational onboarding
- Lovable — UI for both farmer-facing matchmaker and lender-facing discovery portal
- Masumi — payment trigger when farmer selects a product and initiates application
- Python with py2neo or Neo4j Python driver — backend API

✓ Strengths

⚠ Weaknesses

Best technical showcase of the sponsor tech stack — Neo4j is front and centre	Hardest to explain to non-technical judges quickly
Genuinely novel: no such graph-based matchmaking layer exists in Kenyan AgriFin	Graph databases have a learning curve — team needs someone comfortable with Cypher
Serves both farmers and lenders — double-sided platform is a strong business model	Value is only as good as the loan product data seeded into the graph
Graph visualisation in the demo is visually striking and memorable	Risk of the demo looking like a database browser rather than a user product
Strong technical depth appeals to engineering judges	The farmer-facing UX needs careful design so it does not feel like a search engine
Neo4j has its own team as a sponsor — direct positive signal to use it as the core	Without real lender product data, the matches may feel arbitrary

Judge Q&A Simulation — Toughest Questions

These are the questions a judge from Mercy Corps AgriFin, a technical panel member, or a commercial investor would ask. Read every answer carefully — these are where challenges are won or lost.

Q1: "Why use a graph database instead of a simple SQL filter?"

A: Eligibility is not a flat checklist. Whether a farmer qualifies depends on chains of relationships — which crops they grow, which region, which season, what their yield history looks like, who their off-taker is. Graph traversal captures this naturally. SQL would require dozens of joins.

Q2: "How do you get loan product data into the graph?"

A: We manually seeded the graph with 20+ real loan products from publicly available MFI and SACCO product sheets during the preparation phase. AgriFin partners would provide live data feeds post-challenge.

Q3: "Who maintains the graph as products change?"

A: We built an admin API that lenders use to update their own product nodes. Changes propagate instantly to all matches. The graph is designed to be self-maintaining by the lenders themselves.

Q4: "Is this just a comparison website like a financial aggregator?"

A: No. A comparison site shows you all products and lets you filter. Our system understands your specific profile and traverses the relationship graph to return only products you genuinely qualify for, ranked by fit, with an explanation of why.

Q5: "What is your revenue model?"

A: B2B SaaS: lenders pay a subscription to list their products and access farmer leads. Farmers use it free. We can also charge per successful application referral.

Verdict

TECHNICALLY IMPRESSIVE, DEMO RISK — Strong choice if your team has Neo4j experience and a strong designer to make the UX feel like a product, not a database. Risk is that judges may not intuitively feel the value without a polished demo.

PROJECT 04 | AUTONOMOUS LOAN COLLECTION AGENT

"An AI agent that monitors, nudges, and negotiates loan repayments"

Overview

Agricultural loan default rates in Kenya are high not because farmers are unwilling to repay, but because repayment timing is misaligned with harvest cycles, communication is poor, and lenders have no intelligent follow-up system. This project builds an autonomous AI agent that monitors loan repayment status, sends context-aware nudges (harvest season = optimal repayment time), negotiates restructuring when farmers are in difficulty, and uses Masumi's payment rails to actually execute transactions within the conversation.

The Problem It Solves

Non-performing loans (NPLs) in agricultural finance are a systemic problem. Mercy Corps AgriFin partners lose significant capital to defaults that could have been prevented with smarter, timelier intervention. The collection process is manual, expensive, and often adversarial — when it should be supportive and data-driven.

How It Works — Step by Step

16. Loan monitoring layer: connects to lender systems to track repayment status, overdue amounts, and farmer harvest calendar data.
17. Contextual nudge engine: cross-references current date against crop harvest calendars and weather data to identify optimal moments to contact farmers. Sends WhatsApp/SMS messages timed to when farmers actually have cash.
18. Negotiation agent: if a farmer responds indicating difficulty, the AI agent engages in a structured negotiation — offering restructuring options within pre-approved lender parameters.
19. Masumi payment integration: farmer can confirm repayment and execute the M-Pesa transaction directly within the conversation — no need to go to a bank branch or navigate USSD separately.
20. Lender analytics dashboard: shows portfolio health, upcoming harvest-triggered collection opportunities, and agent negotiation outcomes.

Tech Stack & Sponsor Tool Usage

- Featherless LLMs — conversational negotiation and nudge message generation
- Masumi — core payment execution layer; this is Masumi's primary use case
- Neo4j — graph of farmer-loan-lender relationships and repayment history
- Lovable — lender analytics dashboard UI
- WhatsApp Business API or SMS gateway — real-world communication channel

✓ Strengths	⚠ Weaknesses
Masumi is used as the core, not an add-on — strongest alignment with sponsor tools	Ethically sensitive: "debt collection AI" can feel predatory if not framed carefully
Solves a high-value problem for lenders: NPL reduction has direct financial ROI	Masumi integration is the most complex technical piece — payment agent flows are non-trivial
Harvest-aware timing is a genuinely novel and defensible insight	Requires lender system integration (loan data API) which is hard to mock convincingly
Lenders are the paying customer — B2B SaaS is a cleaner revenue model than B2C	WhatsApp Business API has a signup/approval process — cannot be completed during the sprint
Most commercially viable project of the four: lenders will pay real money for this	Risk of over-engineering: the agent architecture is complex and can break during demos
Strong narrative: "We do not chase farmers, we time interventions intelligently"	Judges from a farmer-welfare background (Mercy Corps) may push back on the collection framing

Judge Q&A Simulation — Toughest Questions

These are the questions a judge from Mercy Corps AgriFin, a technical panel member, or a commercial investor would ask. Read every answer carefully — these are where challenges are won or lost.

Q1: "Is this not just a debt collection tool? That sounds predatory."

A: The design philosophy is the opposite of traditional debt collection. We do not chase or pressure. We time outreach to harvest moments when farmers have liquidity, offer restructuring proactively, and the agent is designed to find solutions, not extract payments. Think of it as a financial health check-in, not a collection call.

Q2: "What happens if the AI negotiates a deal the lender has not approved?"

A: The agent operates within a strict parameter set defined by the lender — maximum restructuring terms, minimum acceptable payments, escalation triggers. It cannot offer anything outside those rails without human approval. The lender always has final authority.

Q3: "How does Masumi actually work in this context?"

A: Masumi is a payment network for AI agents. Our agent, once a repayment agreement is reached in conversation, generates a Masumi-signed payment instruction that triggers an M-Pesa transaction. The farmer confirms once; Masumi handles the rest.

Q4: "What about data privacy? Farmer financial data is sensitive."

A: All data is held by the lender, not by us. We process it under a data processing agreement. We comply with Kenya's Data Protection Act 2019. Farmers consent to communication at loan origination.

Q5: "Can this work for farmers who are not smartphone users?"

A: The SMS fallback channel works on any phone. The conversational agent is designed to work over plain SMS as well as WhatsApp, with graceful degradation of features.

Verdict

HIGHEST COMMERCIAL VALUE, HIGHEST RISK — Best choice if your team is strong on backend/agent architecture and you are confident in the Masumi integration. Needs very careful ethical framing. High ceiling, but the most that can go wrong during a demo.

Final Decision Guide

Use this table to match the project to your team's actual strengths. Be honest.

If your team is strongest in...	Choose this project
Data science, ML, graph databases (Neo4j Cypher)	PROJECT 01 — Credit Scoring Agent
Conversational AI, NLP, Swahili language, mobile UX	PROJECT 02 — Input Financing Assistant
Graph architecture, product thinking, two-sided markets	PROJECT 03 — Farmer-Lender Graph
Agent systems, payment APIs, backend architecture, B2B	PROJECT 04 — Loan Collection Agent
Mixed team, no clear specialisation	PROJECT 02 — Lowest technical risk, highest judge appeal

Whichever project you choose: apply now at kenyaaichallenge.com, workshops begin 15 May 2026, and the final sprint is 27–28 June 2026 in Nairobi.